Abhinandan Pandey

LLM Security Researcher — Code Security & Agentic AI Red Teaming
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Summary

AI & Application Security Researcher specializing in LLM code security, multi-turn agentic trajectory evaluation, and MCP vulnerability analysis. Creator of open-source static analysis tooling (`mcp-shield`) for detecting tool poisoning, schema drift, and confused-deputy flaws. Experienced in Python, Java, secure diff review (OWASP Top 10 / CWE Top 25), and deterministic `pytest` security regression tests that verify genuine root-cause remediation.

Technical Skills

Application & AI Security: Agentic Trajectory & Diff Review, MCP & Tool-Calling Security, Prompt/Metadata Injection, Confused Deputy Attacks, OWASP Top 10, CWE/SANS Top 25 (Injection, Path Traversal, Deserialization, Access Control), SAST, Burp Suite, Malware Analysis, CTF (Hacker101)
Engineering & Verification: Python, Java, FastAPI, Docker/Containers, Git, `pytest` (Security Regression & Verifier Testing), Linux, JSON/JSONL
LLM & Evaluation Stack: Multi-Turn Agent Benchmarking, RAG (LangChain, Chroma, FAISS), LoRA Fine-Tuning (Qwen2.5), Ollama, Streamlit

Experience

Independent LLM & Code Security Researcher Jan 2026 – Present

- Evaluate multi-turn LLM coding and tool-calling trajectories in Python and containerised environments to verify patches fix the true root cause without regressions.
- Identify reward-hacking and illegitimate passes in agent runs: test-harness manipulation, PoC overfitting, and suppressed exceptions.
- Author and repair `pytest` security regression suites and static verifiers that fail on vulnerable builds and pass across valid fixes.
- Offensive research across C2 simulation, web exploit labs (injection, SSRF, path traversal, auth bypass), and agentic pipeline red teaming.

Technical Internships
CodeAlpha (Cybersecurity & ML Intern) — **C-DAC / IIT Roorkee** (Quantum Computing Intern) — **CodeSpaze** (App Development Intern) — **CodSoft** (Java Development Intern)

Security Tooling & Open-Source Projects

`mcp-shield` & MCP Security Analyzer — github.com/abhinandanpandey-in/mcp-shield 2026

- Built a working stdio MCP client and open-source static scanner from scratch to audit tool-calling manifests and server schemas.
- Engineered static rules (including MVT-02, MVT-06, MVT-09) to catch tool poisoning, over-permissive schemas, and cross-tool confused-deputy escalation.
- Implemented cryptographic lockfile verification to detect silent rug-pull schema drift; validated rules against vulnerable and patched fixtures with `pytest`.

Threat-Intel RAG & Vulnerability Evaluation Pipeline 2026

- LangChain + Chroma pipeline over the CISA KEV catalog with a LoRA fine-tuned Qwen2.5 model to analyze CVE root causes and remediation diffs.

SecureDL – Security-Hardened Containerised Web Application 2026

- Security-hardened FastAPI + Docker wrapper for `yt-dlp` with input sanitization, command-injection prevention, and isolated execution.

Offline Healthcare RAG Chatbot 2025

- Air-gapped inference pipeline using FastAPI, FAISS, Phi-3 via Ollama, and Streamlit.

Education & Certifications

B.Tech in Computer Science & Engineering – AKTU / Lucknow Institute of Technology **CGPA: 7.7** — 2026
Certifications & Training: Certified Red Team Operations Management (CRTOM) — OSCP (In Progress) — 1-Year Cybersecurity Diploma (Craw Security, Saket) — Malware Analysis (Red Team Leaders)
Achievements: Solved 1,000+ Data Structures & Algorithms (DSA) problems; Active CTF player (Hacker101)